



Long-Term Consequences of Teaching Gender Roles

Evidence from Desegregating Industrial Arts and Home Economics in Japan

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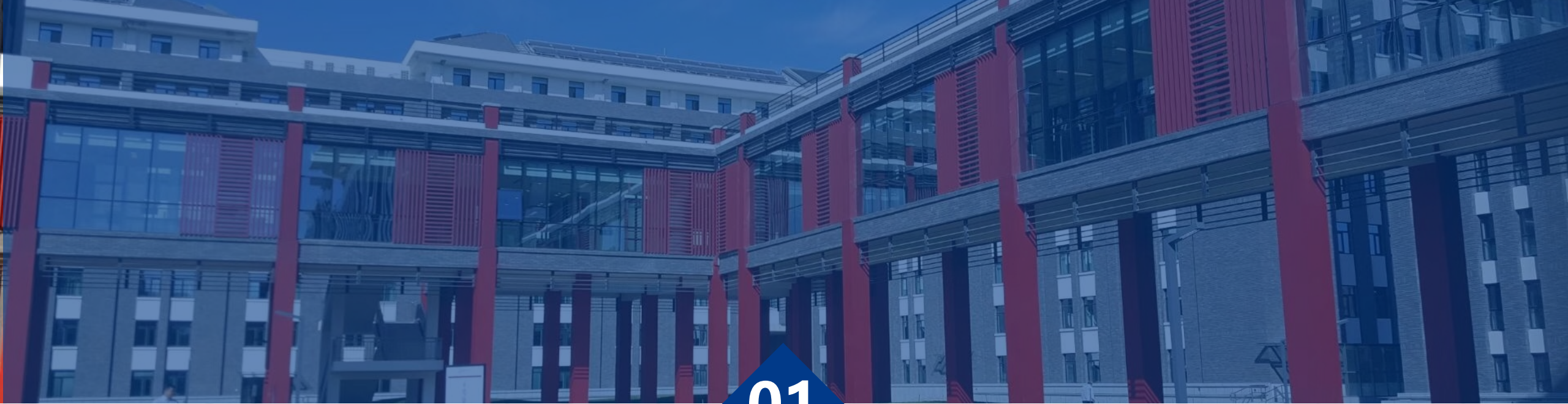
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结论

A photograph of a modern building with a courtyard, featuring prominent red vertical pillars and glass facades. The image is overlaid with a blue tint.

01

Introduction & Motivation

Authors lay down the situation of Japanese education prior to the reform and the factors that triggered the change

The Macro Puzzle & Gender Asymmetry in Japan

High FLFP, Low Permanent Roles

Female labor force participation has crossed 70%, yet only 44% hold full-time permanent roles, contributing to a persistent gender wage gap.

The Home Front

Japanese husbands with young kids spend just 1 hour 23 minutes daily on domestic work—the lowest in the OECD.

Normative Constraints

Strong traditional beliefs persist; historically, 83% agreed a wife earning more than her husband causes domestic friction.

Gender Asymmetry Framework

1. Labor Market

Women experience high participation rates (>70%) but low rates of permanent, full-time employment (44%).

2. Household Division

Japanese men spend the least time on domestic work among OECD nations (~1.4 hours/day for fathers).

3. Social Norms

Traditional values remain strong; a large majority historically viewed female breadwinners as a source of conflict.

A photograph of a modern, multi-story building with a curved facade. The building features large glass windows and prominent red vertical columns. The image is overlaid with a blue tint. Below the photograph is a white diamond shape containing the number 02.

02

Institutional Background & The Policy Shock

To answer our question, the authors analyze a major educational reform that dramatically altered how gender roles were taught in Japanese schools.

The Evolution of Home Economics (1950–1990)



1990 MEXT Reform

The gender-segregated framework was abolished, establishing a unified, coeducational curriculum.

1978 GGE Revision

"Pseudo-Choice" era where boys took 5 IA + 1 HE, and girls took 5 HE + 1 IA, maintaining 83% content differentiation.

1958 GGE Revision

Split into "Industrial Arts" (IA) for boys and "Home Economics" (HE) for girls, creating strict gender segregation.

1947–1950 Post-War Genesis

A single, coeducational course for all children.

04

03

02

01

Key
Takeaway

This timeline highlights the gradual shift from enforced gender roles to equality in education.

The 1990 MEXT Educational Reform



Exogenous Trigger

Political urgency to ratify the UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), which prompted educational policy adjustments.

The Shift: Unified Coeducation

- Abolished segregated tracks on **April 1, 1990**.
- Introduced a mandatory unified curriculum.
- Boys and girls shared classrooms and identical hours of both Industrial Arts (IA) and Home Economics (HE).

Treatment Cohort Discontinuity

- The "1977 cohort" (born Apr 1977 - Mar 1978) was the first to experience **all 3 years** of the unified curriculum.
- Clear policy cutoff: Those born before experienced the old system; those after the new integrated one.

"This policy discontinuity is the core of the identification strategy to study the reform's impact."

A photograph of a modern, multi-story building with a curved facade. The building features large glass windows and prominent red vertical columns. The image is overlaid with a blue tint.

03

Methodology

Research Design & Approach

Authors use a Regression Discontinuity (RD) design to isolate the causal effect of the 1990 reform on adult outcomes measured 26 years later.

Regression Discontinuity (RD) Design

01. Identification Strategy

Sharp RD taking advantage of the deterministic boundary dictated by an individual's academic birth cohort.

02. Econometric Specification

$$Y_i = \alpha + \beta \text{Post}_{1977} + \delta X_i + \sum \mu_k(\pi_j) + \sum \eta_m(Q_m) + \text{Trends} + \varepsilon_i$$

s.t. Trends = $(1 - \text{Post}_{1977}) f_0(D_i) + (\text{Post}_{1977}) f_1(D_i)$

03. Running Variable

Academic year of birth, normalized to 0 at the April 1977 cutoff.

04. Key Parameter (β)

Measures the long-term causal effect of the coeducational shift on adult behaviors.

Data & Identification Validity

Valid Data

JTUS 2016
15-min precision

Appropriate Sample

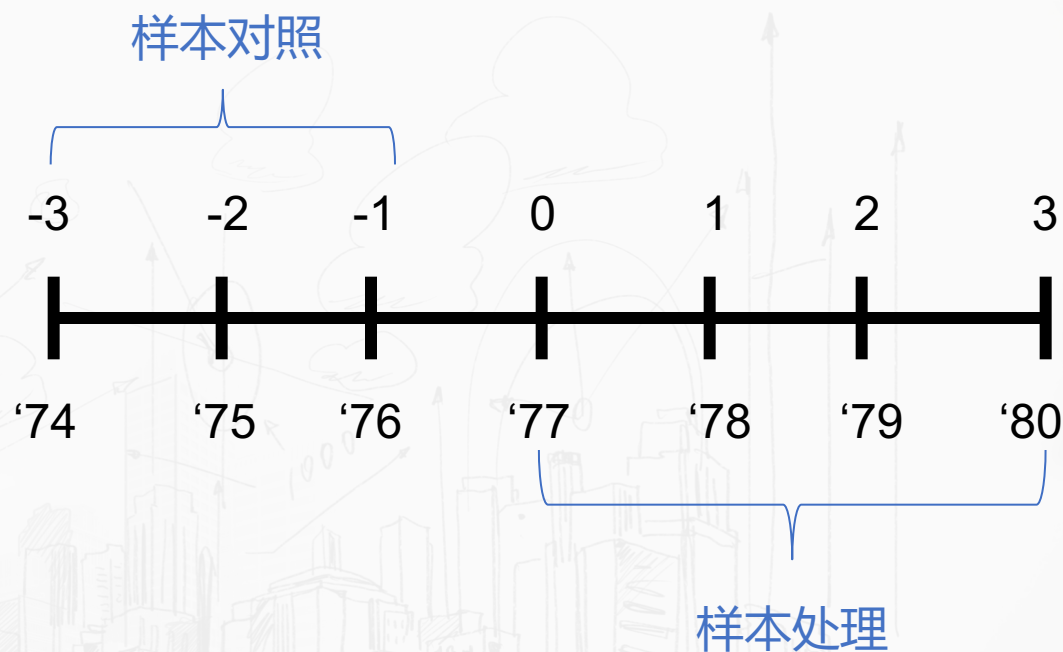
Married, Age 37-42
Established Families
5,393 men
6,251 women

No Manipulation

Smooth distribution
of birth dates

Balanced Groups

Similar characteristics
across threshold



A photograph of a modern, multi-story building with a curved facade. The building features large glass windows and prominent red vertical columns. The sky is clear and blue.

04

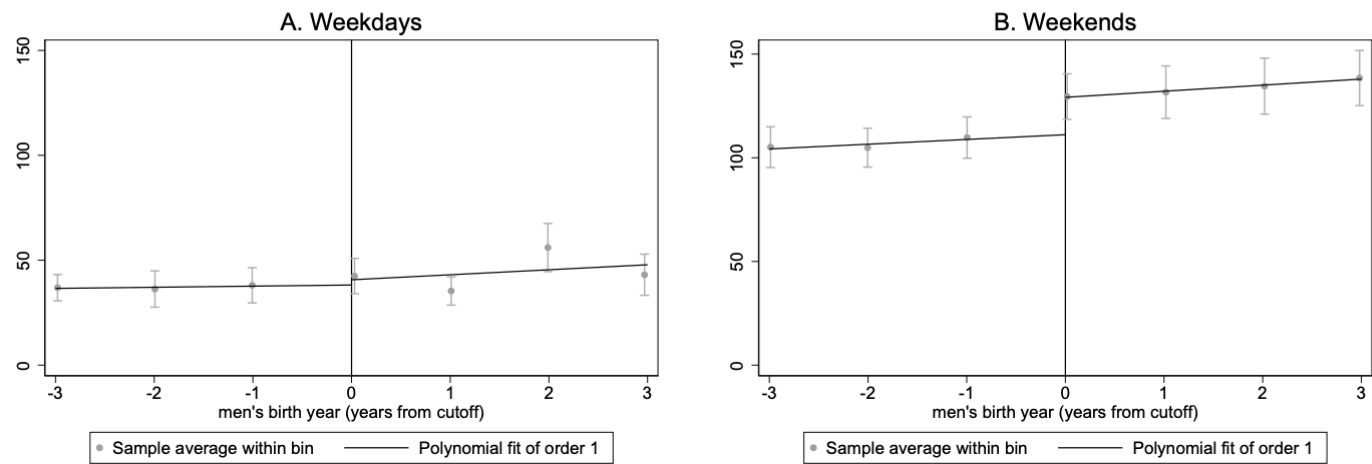
Findings

WEEKEND AND INDIRECT EFFECTS ON MEN

The reform had profound and lasting effects on both household dynamics and labor market outcomes.

Figure 2A: Home Production Time, by Gender

(a) Men



(b) Women

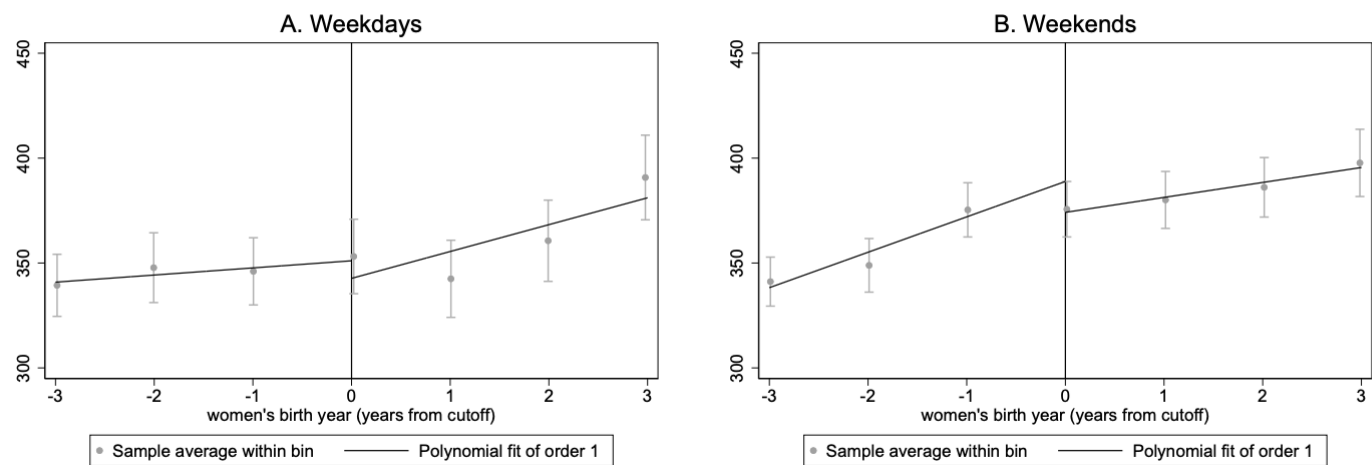
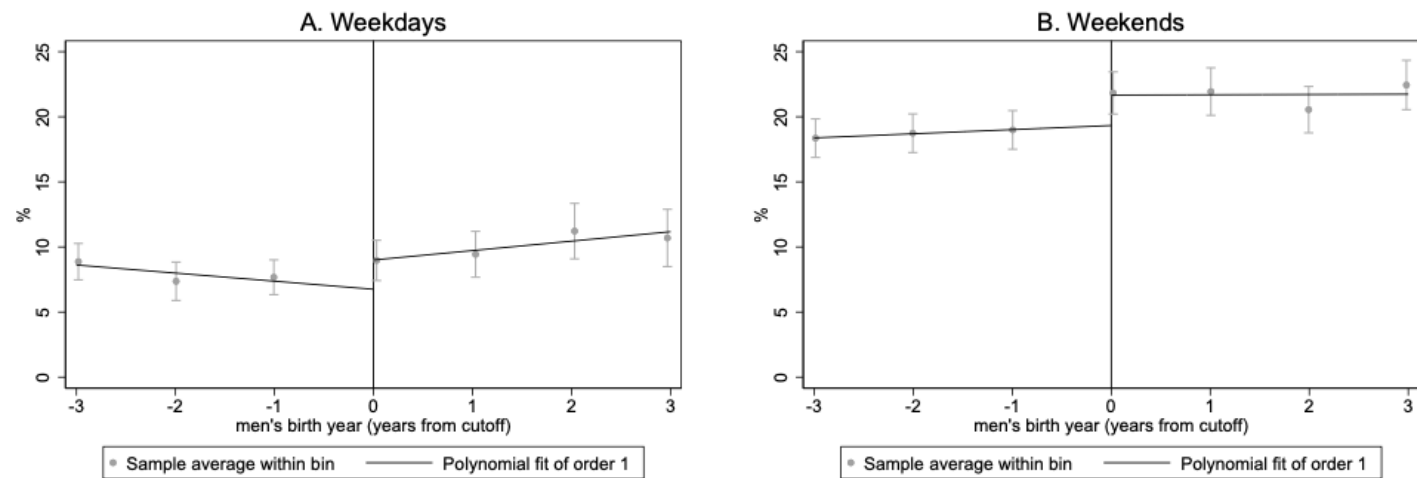


Figure 2B: Share of Home Production within A Couple, by Gender (Couple Sample)

(a) Men



(b) Women

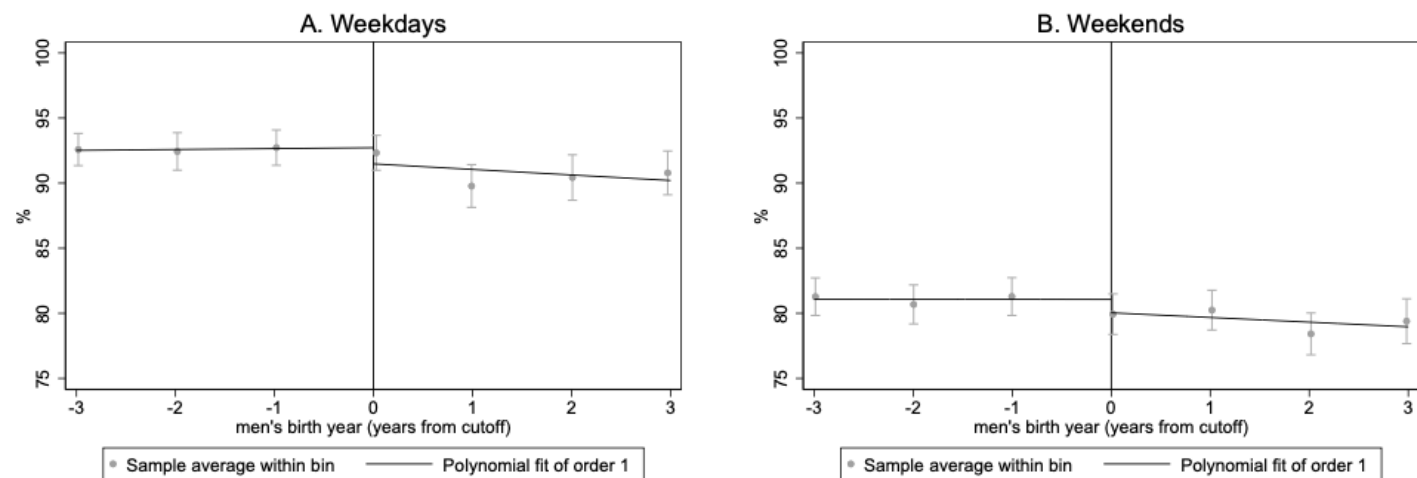


Table 4. Home-Production Time

	Spec 1	Spec 2	Spec 3
<i>Panel A: Men</i>			
<u>Weekdays</u>			
Time (min)	0.437 (2.727)	0.477 (2.809)	-2.271 (3.210)
Share relative to spouse (%)	1.950** (0.603)	1.536** (0.551)	1.382** (0.557)
<u>Weekends</u>			
Time (min)	19.653*** (1.205)	24.129*** (3.462)	20.481*** (2.522)
Share relative to spouse (%)	2.389*** (0.298)	2.362*** (0.233)	2.012*** (0.259)
<i>Panel B: Women</i>			
<u>Weekdays</u>			
Time (min)	-10.238 (11.609)	-9.310 (13.193)	-17.852 (12.900)
Share relative to spouse (%)	-0.831 (0.850)	-0.999 (0.793)	-0.943 (0.814)
<u>Weekends</u>			
Time (min)	-15.942** (5.426)	-9.773 (5.973)	-7.226 (5.861)
Share relative to spouse (%)	-1.293** (0.481)	-1.319*** (0.318)	-1.003** (0.385)
Years of education		X	X
Three-generation household		X	X
Total number of children			X
Number of children under 10			X

Notes: Estimates are from a 3-year bandwidth RD model with dummy variables for prefecture and day of the week and additional covariates when indicated in the bottom rows of the table. Robust standard errors are clustered at the year of birth level. Sample sizes of time are 3,564 for men and 4,589 for women on weekdays and 6,371 for men and 7,712 for women on weekends. Share relative to spouse are estimated using a sample that includes his/her own spouse; therefore, sample sizes are 3,289 for men and 3,739 for women on weekdays, and 5,764 for men and 6,546 for women on weekends. The symbols ***, **, and * indicate that the pre/post-reform differences are statistically significant at the 1%, 5%, and 10% level, respectively.

Figure 5: Labor Market Outcome, Women

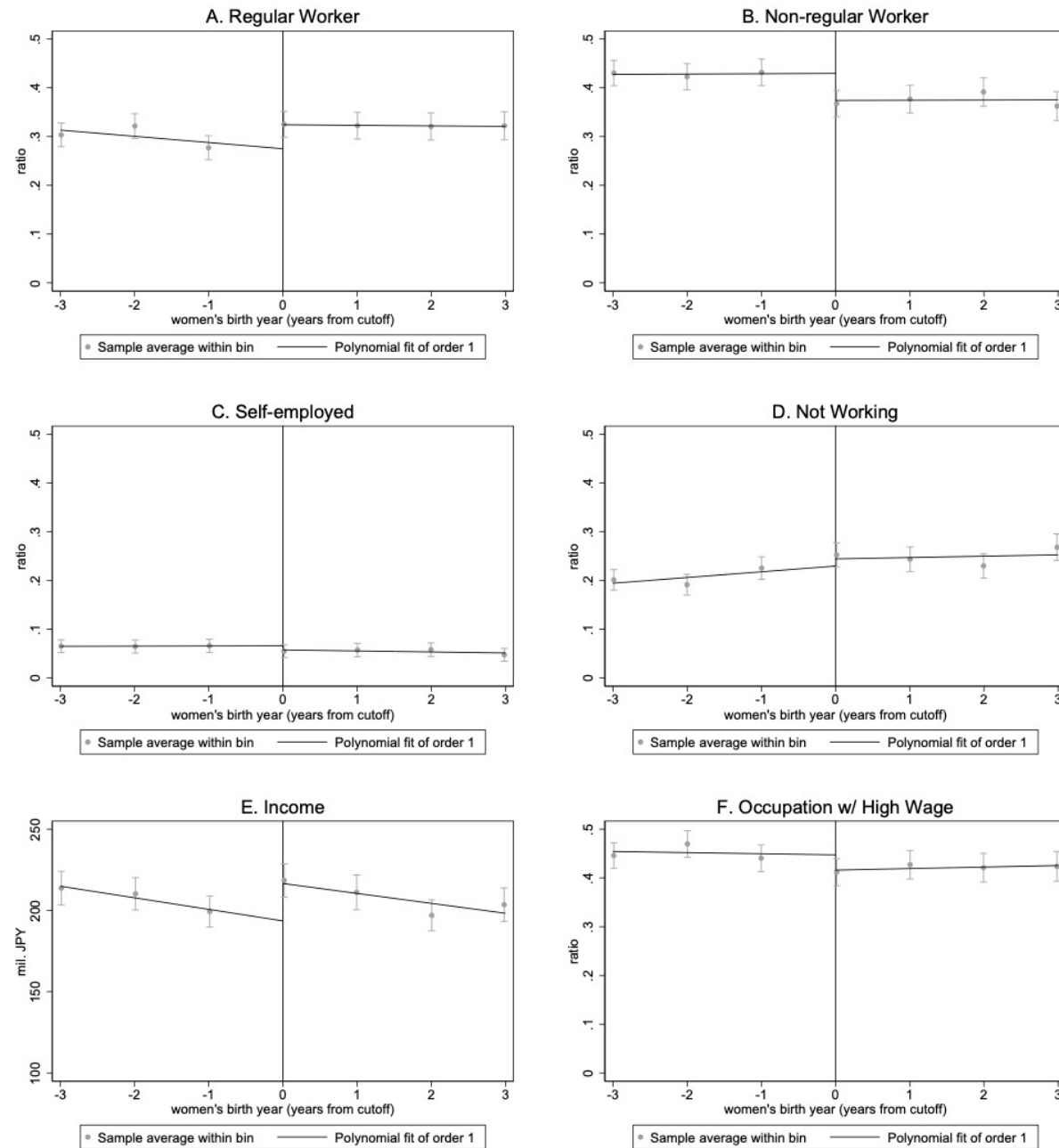


Table 6. Labor Market Outcomes

Regular employment	Non-regular employment	Self-employed	Not working	Annual income	High-wage occupation	Weekday time spent working (in minutes)
Men						
-0.006 (0.011)	0.012 (0.008)	-0.001 (0.008)	-0.004 (0.004)	-0.951 (11.875)	0.013 (0.015)	-4.737 (7.570)
Women						
0.047** (0.017)	-0.056*** (0.009)	-0.015* (0.007)	0.024 (0.023)	23.641*** (2.237)	-0.025 (0.018)	6.999 (16.808)

Notes: Estimates are from a 3-year bandwidth RD model with dummy variables for prefecture and day of the week. Robust standard errors are clustered at the year of birth level. The sample for labor market outcomes is 5,393 for men (5,311 for annual income) and 6,251 for women (4,582 for annual income). The unit for annual income is 10,000 Japanese yen (\$1 = 107JPY). Sample sizes of work-related time on weekdays are 3,564 for men and 4,589 for women. The symbols ***, **, and * indicate that the pre/ post-reform differences are statistically significant at the 1%, 5%, and 10% level, respectively.

Table 9. Women's Labor-Market Outcomes Within a Couple

	Women's Treatment Running Variable: Women's Date of Birth		Men's Treatment Running Variable: Men's Date of Birth	
	(1)	(2)	(3)	(4)
	Panel A <i>Direct Effect on Women Themselves</i>		Panel B <i>Indirect Effect on their Wife</i>	
Sub-sample	Husband also treated	Husband untreated	Wife also treated	Wife untreated
Regular employment	0.178*** (0.025)	0.061** (0.009)	-0.017 (0.44)	-0.03 (0.031)
Non-regular employment	-0.184*** (0.039)	-0.054*** (0.013)	0.044 (0.30)	-0.049 (0.026)
Self-employed	-0.058*** (0.009)	-0.014 (0.007)	-0.011 (0.013)	0.030** (0.011)
Not working	0.064 (0.061)	0.007 (0.008)	-0.016 (0.016)	0.049 (0.041)
Annual income (unit: 10,000 JPY)	38.461*** (5.889)	23.227** (6.299)	11.653 (9.300)	-5.773 (6.348)
High-wage Occupation	0.048 (26.53)	0.001 (0.009)	-0.005 (0.025)	-0.106*** (0.018)
Number of observations	1,682	1,886	2,121	1,447

Source: Statistics Bureau of Japan, 2016 JTUS.

Notes: Estimates are from a 3-year bandwidth RD model with dummy variables for prefecture and day of the week. Robust standard errors are clustered at the year of birth level. Only a sample which includes his/her own spouse is used. The forcing variable for both husbands and wives is restricted to $(fv \geq -3 \ \& \ fv \leq 3)$, meaning that all samples were born between 1974 and 1980. Sample sizes of weekday time spent working are 1,097 and 1,204 for treated women with treated husband and untreated husband, respectively, and are 1,366 and 935 for their wives with treated wife and untreated wife, respectively. The symbols ***, **, and * indicate that the pre/post-reform differences are statistically significant at the 1%, 5%, and 10% level, respectively.

Table 10. Weekend Home-Production Time Within a Couple (in minutes)

	Women's Treatment Running Variable: Women's Date of Birth		Men's Treatment Running Variable: Men's Date of Birth	
	(1)	(2)	(3)	(4)
	Panel A <i>Indirect Effect on their Husband</i>		Panel B <i>Direct Effect on Men Themselves</i>	
Sub-sample	Husband also treated	Husband untreated	Wife also treated	Wife untreated
Home Prod.	45.23 (23.85)	32.32*** (6.59)	-3.55 (5.85)	27.99*** (6.01)
Housework	11.63*** (2.56)	-2.34 (5.96)	-5.76** (1.78)	-4.55 (4.18)
Childcare	8.79 (19.17)	20.61*** (3.632)	8.42 (5.15)	28.38*** (6.38)
Other	24.81* (7.20)	14.05*** (2.601)	11.64 (9.97)	4.15 (3.55)
Leisure	18.52* (9.18)	-15.02* (7.21)	-41.53** (11.94)	26.53* (13.03)
Life-Support	-32.51*** (8.25)	5.08 (3.57)	21.26* (8.77)	-9.94 (10.46)
Work-Related	-31.24 (18.87)	-22.39** (6.20)	23.81** (8.29)	-44.57*** (10.97)
	Panel C <i>Direct Effect on Women Themselves</i>		Panel D <i>Indirect Effect on Their Wives</i>	
Sub-sample	Husband also treated	Husband untreated	Wife also treated	Wife untreated
Home Prod.	-58.77 (32.39)	-18.28 (10.08)	-2.88 (14.61)	-19.79 (28.90)
Housework	-18.57 (11.14)	0.13 (8.13)	-28.88*** (7.21)	-11.07 (7.78)
Childcare	-48.16** (13.17)	-4.77 (5.54)	26.88*** (2.40)	5.29 (26.17)
Other	7.86 (6.69)	-13.64** (4.13)	-0.87 (8.33)	-14.01*** (3.38)
Leisure	2.17 (15.83)	0.538 (8.73)	-14.52 (14.99)	-2.20 (16.89)
Life-Support	0.67 (15.83)	0.123 (1.54)	-2.49 (4.71)	1.625 (13.23)
Work-Related	55.93* (26.53)	17.62*** (2.29)	19.89* (8.32)	20.36 (25.71)
Number of observations	1,844	2,171	2,368	1,647

Notes: Estimates are from a 3-year bandwidth RD model with dummy variables for prefecture and day of the week. Robust standard errors are clustered at the year of birth level. Only a sample which includes his/her own spouse is used. The forcing variable for both husbands and wives is restricted to (fv>= -3 & fv<= 3), meaning that all samples were born between 1974 and 1980. The symbols ***, **, and * indicate that the pre-/post-reform differences are statistically significant at the 1%, 5%, and 10% level, respectively.

Re-allocating Time Within the Household (Weekend Effects)



北京师范大学
BEIJING NORMAL UNIVERSITY

Treated Men

Increased weekend home production by **20 minutes/day (+18%)** and boosted their relative share of domestic labor by **2.3 percentage points (+13%)**.

Treated Women

Reduced weekend home production by **16 minutes (-5%)** and lowered their overall domestic share of household chores.

The Trade-off

A clean structural shift on weekends where gender gaps in domestic labor and paid market work narrow concurrently, indicating a more equitable division.

Weekday vs. Weekend

The effects are concentrated on weekends, suggesting that the response is primarily to the structure of the labor market and free time availability.

家庭时间分配
(Household Time Allocation)

Spillovers to the Labor Market

Women's Employment

Treated women see a **5 percentage point (+20%) surge** in the probability of holding regular, permanent employment.

Men's Employment

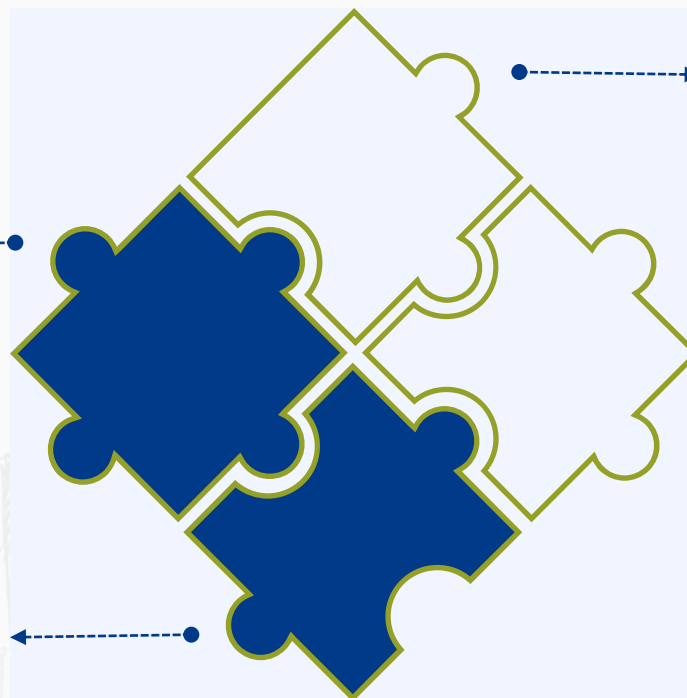
Precise **zero effects** on men's employment status.

Women's Income

Treated women experience a **5% increase** in annual labor income.

Men's Wages

Precise **zero effects** on men's wages.



The reform systematically shrinks the gender wage and employment gap.

The Spousal Treatment Matrix

Authors separate outcomes based on whether the individual's spouse was also treated or untreated.

Women's Primacy

The reform acts as a primary **direct shock** to women's labor supply and household bargaining.

Men's Passivity

The expansion of men's home production is predominantly an **indirect effect**—a response to their wives' heightened labor market commitment. Results persist regardless of children.



Mechanism of Change



05

Conclusion & Policy Discussion

Authors explore the role of changing norms versus skill accumulation and verify the robustness of our findings.

Mechanisms: Shifting Norms vs. Pure Skill Accumulation

Skill Accumulation

Students learn non-traditional skills (e.g., boys learn cooking, girls learn mechanics).

Attitudinal Shifts

Treated women show a statistically significant increase in rejecting traditional gender norms.

Fertility Dynamics

The reform delayed fertility overall but accelerated it when childcare burdens were reduced, further reinforcing the norm shift.

Behavioral Change

This leads to increased female labor supply and renegotiation of household tasks.



Key Policy Takeaways



Durable Transformations

Early institutional curriculum shifts can cause deep, durable behavioral transformations that survive well into mid-adulthood.



Identity as a Lever

Altering female gender identity acts as a major lever for shifting intra-household labor allocations and reducing gender inequality.



Labor Market Bottleneck

The impact is concentrated on weekends because of Japan's demanding regular employment system. Educational changes must be met with labor-market structure updates to maximize their full potential.



Thank You

Q & A Session